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CYBERSTER PURPLE TEAM INTERNSHIP

Batch 1 — Month 1: Network Security & Protocol Analysis

WEEK 2 TECHNICAL REPORT

Network Discovery & Reconnaissance

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Report Date	24 May 2026
Week	Week 2 of 12
Module	Month 1: Network Security & Protocol Analysis

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Objective

To contrast aggressive scanning techniques with stealthy discovery and implement the first layer of defensive detection. This assignment covers both the attacker's perspective (active and passive reconnaissance) and the defender's perspective (IDS implementation and alert analysis).

Task 01: Active & Passive Discovery

This task involved mapping the network using both loud and silent methodologies, comparing scanning techniques, and performing service fingerprinting.

1.1 Reconnaissance Methods: Active vs Passive

Passive Reconnaissance

Passive reconnaissance involves collecting information about a target without directly interacting with it. The target has no awareness that information is being gathered. This method is stealthy and leaves no traces in server logs.

- Searching domain registration data via whois
- DNS enumeration using nslookup
- Searching public databases and OSINT platforms
- Analyzing social media and publicly available documents

Active Reconnaissance

Active reconnaissance involves directly interacting with the target system to gather information. Because packets are sent to the target, it may detect or log the activity.

- Port scanning with Nmap
- Service version detection (-sV flag)
- ARP/ICMP sweeps to identify live hosts
- TCP SYN scanning

Why Reconnaissance is Important in Pentesting

Reconnaissance is the foundation of any penetration test. Without it, a pentester is operating blindly. By understanding the target's attack surface — open ports, running services, OS versions, and domain structure — a pentester can identify potential vulnerabilities before attempting exploitation. Skipping this phase dramatically increases the risk of missing critical entry points or triggering unnecessary alerts.

1.2 DNS & Whois Enumeration on example.com

Whois Enumeration

```
whois example.com
```

Results showed the domain is registered under IANA (Internet Assigned Numbers Authority), created in 1992, with Cloudflare nameservers (ELLIOTT.NS.CLOUDFLARE.COM and HERA.NS.CLOUDFLARE.COM).

```

kali@kali: ~/Desktop
Session Actions Edit View Help
Services: ("VeriSign") Whois database is provided by VeriSign for
information purposes only, and to assist persons in obtaining information
about or related to a domain name registration record. VeriSign does not
guarantee its accuracy. By submitting a Whois query, you agree to abide
by the following terms of use: You agree that you may use this Data only
for lawful purposes and that under no circumstances will you use this Data
to: (1) allow, enable, or otherwise support the transmission of mass
unsolicited, commercial advertising or solicitations via e-mail, telephone,
or facsimile; or (2) enable high volume, automated, electronic processes
that apply to VeriSign (or its computer systems). The compilation,
repackaging, dissemination or other use of this Data is expressly
prohibited without the prior written consent of VeriSign. You agree not to
use electronic processes that are automated and high-volume to access or
query the Whois database except as reasonably necessary to register
domain names or modify existing registrations. VeriSign reserves the right
to restrict your access to the Whois database in its sole discretion to ensure
operational stability. VeriSign may restrict or terminate your access to the
Whois database for failure to abide by these terms of use. VeriSign
reserves the right to modify these terms at any time.

The Registry database contains ONLY .COM, .NET, .EDU domains and
Registrars.
% IANA WHOIS server
% for more information on IANA, visit http://www.iana.org
% This query returned 1 object

domain:      EXAMPLE.COM
organisation: Internet Assigned Numbers Authority
created:     1992-01-01
source:      IANA

(kali@kali) (~/Desktop)
$

```

Figure 1: whois example.com — domain registration details

DNS AAAA Record (IPv6)

```
nslookup -type=AAAA example.com
```

Returned IPv6 addresses: 2606:4700::6812:1b78 and 2606:4700::6812:1a78 — both Cloudflare CDN addresses.

```

kali@kali: ~/Desktop
Session Actions Edit View Help
$ cd ~/Desktop
(kali@kali) (~/Desktop)
$ nslookup example.com
Domain Name: EXAMPLE.COM
Registry Domain ID: 2336799_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.iana.org
Registrar URL: http://res-dom.iana.org
Updated Date: 2026-01-16T15:25:50Z
Creation Date: 1995-08-14T04:00:00Z
Registry Expiry Date: 2026-08-13T04:00:00Z
Registrar: RESERVED-Internet Assigned Numbers Authority
Registrar IANA ID: 376
Registrar Abuse Contact Email:
Registrar Abuse Contact Phone:
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhib
ited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferPr
ohibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhib
ited
Name Server: ELLIOTT_NS.CLOUDFLARE.COM
Name Server: HERA_NS.CLOUDFLARE.COM
DNSSEC: signedDelegation
DNSSEC DS Data: 2371 13 2 C988EC423E3880EB8D08A46FE06CA230EE23F35B578D64E78B29
C3E1C83D245A
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-03-15T15:29:09Z <<<

For more information on Whois status codes, please visit https://icann.org/epp

NOTICE: The expiration date displayed in this record is the date the
registrar's sponsorship of the domain name registration in the registry is
currently set to expire. This date does not necessarily reflect the expiration
date of the domain name registrant's agreement with the sponsoring
registrar. Users may consult the sponsoring registrar's Whois database to
64 bytes from 10.0.2.2: icmp_seq=62 ttl=255 time=0.528 ms
64 bytes from 10.0.2.2: icmp_seq=63 ttl=255 time=0.646 ms
64 bytes from 10.0.2.2: icmp_seq=64 ttl=255 time=0.478 ms
64 bytes from 10.0.2.2: icmp_seq=65 ttl=255 time=0.435 ms
64 bytes from 10.0.2.2: icmp_seq=66 ttl=255 time=0.720 ms
64 bytes from 10.0.2.2: icmp_seq=67 ttl=255 time=0.514 ms
64 bytes from 10.0.2.2: icmp_seq=68 ttl=255 time=0.393 ms
64 bytes from 10.0.2.2: icmp_seq=69 ttl=255 time=0.360 ms
64 bytes from 10.0.2.2: icmp_seq=70 ttl=255 time=0.587 ms
64 bytes from 10.0.2.2: icmp_seq=71 ttl=255 time=0.407 ms
64 bytes from 10.0.2.2: icmp_seq=72 ttl=255 time=0.364 ms
64 bytes from 10.0.2.2: icmp_seq=73 ttl=255 time=0.453 ms
64 bytes from 10.0.2.2: icmp_seq=74 ttl=255 time=0.373 ms
64 bytes from 10.0.2.2: icmp_seq=75 ttl=255 time=0.352 ms
64 bytes from 10.0.2.2: icmp_seq=76 ttl=255 time=0.381 ms
64 bytes from 10.0.2.2: icmp_seq=77 ttl=255 time=0.509 ms
64 bytes from 10.0.2.2: icmp_seq=78 ttl=255 time=0.361 ms
64 bytes from 10.0.2.2: icmp_seq=79 ttl=255 time=0.519 ms
64 bytes from 10.0.2.2: icmp_seq=80 ttl=255 time=1.49 ms
64 bytes from 10.0.2.2: icmp_seq=81 ttl=255 time=0.429 ms
64 bytes from 10.0.2.2: icmp_seq=82 ttl=255 time=0.324 ms
^C
-- 10.0.2.2 ping statistics --
82 packets transmitted, 82 received, 0% packet loss, time 81831ms
rtt min/avg/max/mdev = 0.302/0.535/2.416/0.276 ms
(kali@kali) (~/Desktop)
$ ls -la
total 8
drwxr-xr-x 2 kali kali 4096 Mar 8 05:16 .
drwxr-xr-x 16 kali kali 4096 Mar 15 07:41 ..
cat: alert_fast.txt: No such file or directory
(kali@kali) (~/Desktop)
$

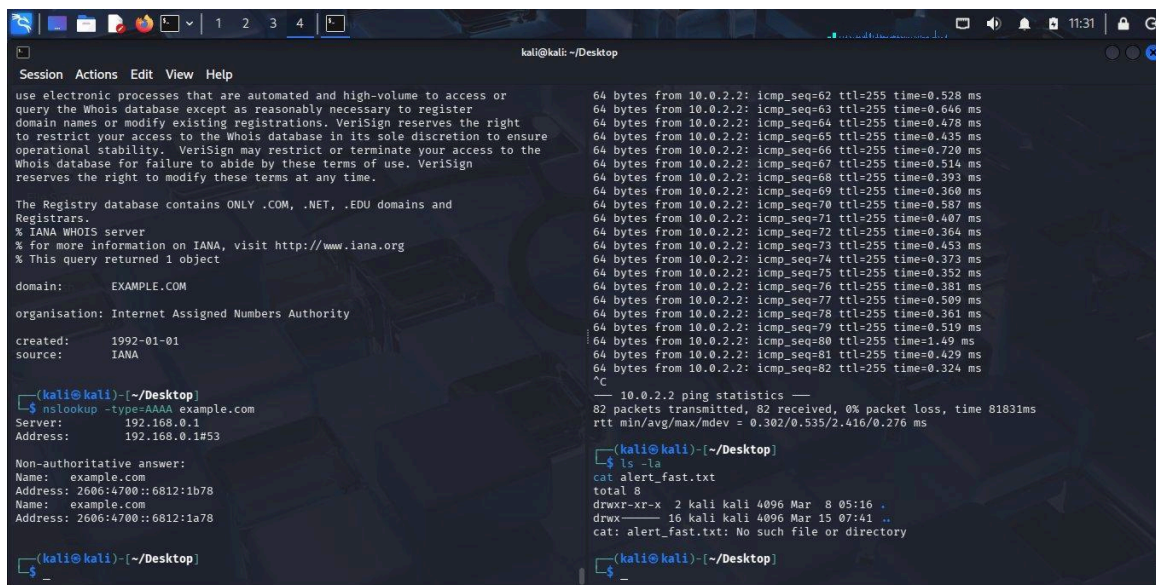
```

Figure 2: nslookup AAAA record — IPv6 addresses for example.com

DNS CNAME Record

```
nslookup -type=CNAME example.com
```

No CNAME record exists — example.com uses A/AAAA records directly. The SOA record shows `elliott.ns.cloudflare.com` as the authoritative nameserver.



```
Session Actions Edit View Help
use electronic processes that are automated and high-volume to access or
query the Whois database except as reasonably necessary to register
domain names or modify existing registrations. VeriSign reserves the right
to restrict your access to the Whois database in its sole discretion to ensure
operational stability. VeriSign may restrict or terminate your access to the
Whois database for failure to abide by these terms of use. VeriSign
reserves the right to modify these terms at any time.

The Registry database contains ONLY .COM, .NET, .EDU domains and
Registrars.
% IANA WHOIS server
% for more information on IANA, visit http://www.iana.org
% This query returned 1 object

domain:      EXAMPLE.COM
organisation: Internet Assigned Numbers Authority
created:     1992-01-01
source:      IANA

(kali@kali) (~/Desktop)
$ nslookup -type=AAAA example.com
Server:      192.168.0.1
Address:     192.168.0.1#53

Non-authoritative answer:
Name:        example.com
Address: 2606:4700::6812:1b78
Name:        example.com
Address: 2606:4700::6812:1a78

(kali@kali) (~/Desktop)
$ nslookup -type=CNAME example.com
Server:      192.168.0.1
Address:     192.168.0.1#53

Non-authoritative answer:
** Can't find example.com: No answer

Authoritative answers can be found from:
example.com
  origin = Elliott.ns.cloudflare.com
  mail addr = dns.cloudflare.com
  serial = 2397268091
  refresh = 10000
  retry = 2400
  expire = 604800
  minimum = 1800

(kali@kali) (~/Desktop)
$ nslookup -type=MX example.com
Server:      192.168.0.1
Address:     192.168.0.1#53

Non-authoritative answer:
example.com  mail exchanger = 0 .

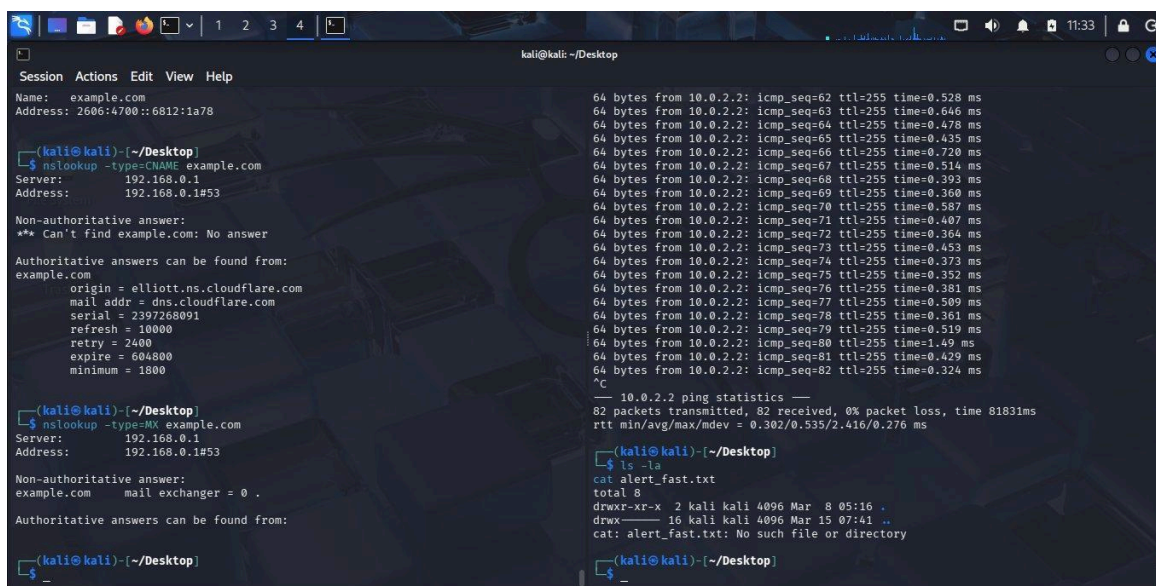
Authoritative answers can be found from:
```

Figure 3: nslookup CNAME record — No answer / SOA record

DNS MX Record (Mail Exchanger)

```
nslookup -type=MX example.com
```

The MX record returned mail exchanger = 0 . — indicating example.com has no dedicated mail server configured.



```
Session Actions Edit View Help
Name:        example.com
Address: 2606:4700::6812:1a78

(kali@kali) (~/Desktop)
$ nslookup -type=CNAME example.com
Server:      192.168.0.1
Address:     192.168.0.1#53

Non-authoritative answer:
** Can't find example.com: No answer

Authoritative answers can be found from:
example.com
  origin = Elliott.ns.cloudflare.com
  mail addr = dns.cloudflare.com
  serial = 2397268091
  refresh = 10000
  retry = 2400
  expire = 604800
  minimum = 1800

(kali@kali) (~/Desktop)
$ nslookup -type=MX example.com
Server:      192.168.0.1
Address:     192.168.0.1#53

Non-authoritative answer:
example.com  mail exchanger = 0 .

Authoritative answers can be found from:
```

Figure 4: nslookup MX record — mail exchanger result

1.3 Stealthy Scan: TCP SYN Scan with Wireshark

```
sudo nmap -sS 10.0.2.2
```

The TCP SYN Scan sends a SYN packet to each port. If the port is open, the target replies with SYN-ACK. The scanner immediately sends RST — never completing the 3-way handshake. This makes the scan harder to log.

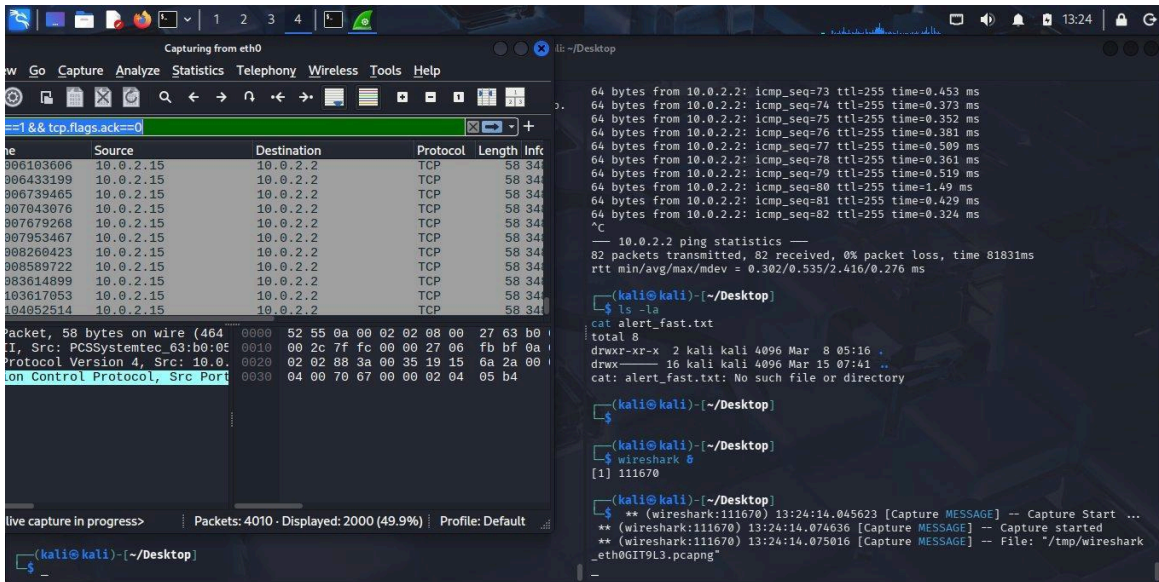


Figure 5: Wireshark capturing TCP SYN packets — filter: `tcp.flags.syn==1 && tcp.flags.ack==0`

1.4 Which Scan to Use for Stealth?

TCP SYN Scan (-sS) is the best choice when stealth is required. Unlike the Full Connect Scan (-sT) which completes the full 3-way handshake and is fully logged, the SYN scan sends RST before the connection is established — many older IDS systems will not log it as a completed connection. It is faster, quieter, and preferred for stealthy penetration testing.

Feature	TCP SYN Scan (-sS)	Full Connect Scan (-sT)
Handshake	Incomplete (SYN→SYN-ACK→RST)	Complete (SYN→SYN-ACK→ACK)
Stealth Level	High — less likely logged	Low — fully logged by target
Speed	Faster	Slower
Root Required	Yes (raw packets)	No
Detection Risk	Low	High

1.5 Service Version Detection with Nmap

```
sudo nmap -sV example.com
```


The `-sV` flag performs service fingerprinting — determining the exact version of software running on each open port. Knowing the exact version allows the tester to search for CVEs and known exploits.

```
kali@kali: ~/Desktop
Session Actions Edit View Help
retry = 2400
expire = 604800
minimum = 1800

(kali@kali) ~/Desktop
$ nslookup -type=MX example.com
Server: 192.168.0.1
Address: 192.168.0.1#53

Non-authoritative answer:
example.com mail exchanger = 0 .

Authoritative answers can be found from:

(kali@kali) ~/Desktop
$ sudo nmap -sV example.com
[sudo] password for kali:
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-15 11:35 EDT
Nmap scan report for example.com (104.18.27.120)
Host is up (0.0080s latency).
Other addresses for example.com (not scanned): 104.18.26.120 2606:4700::6812:1a78
2606:4700::6812:1b78
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
80/tcp    open  tcpwrapped
443/tcp    open  tcpwrapped
8080/tcp   open  tcpwrapped

Service detection performed. Please report any incorrect results at https://nmap.org/submit/.
Nmap done: 1 IP address (1 host up) scanned in 41.74 seconds

(kali@kali) ~/Desktop
$
```

Figure 6: `nmap -sV example.com` — open ports 80, 443, 8080 (tcpwrapped = Cloudflare proxy)

```
kali@kali: ~/Desktop
Session Actions Edit View Help

(kali@kali) ~/Desktop
$ nslookup -type=CNAME example.com
Server: 192.168.0.1
Address: 192.168.0.1#53

Non-authoritative answer:
** Can't find example.com: No answer

Authoritative answers can be found from:
example.com
origin = elliottns.cloudflare.com
mail addr = dns.cloudflare.com
serial = 2397268091
refresh = 10000
retry = 2400
expire = 604800
minimum = 1800

(kali@kali) ~/Desktop
$ nslookup -type=MX example.com
Server: 192.168.0.1
Address: 192.168.0.1#53

Non-authoritative answer:
example.com mail exchanger = 0 .

Authoritative answers can be found from:

(kali@kali) ~/Desktop
$ sudo nmap -sS 10.0.2.2
[sudo] password for kali:
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-15 11:35 EDT
Nmap scan report for 10.0.2.2
Host is up (0.0000s latency).
Not shown: 65535 closed tcp ports (reset)
PORT      STATE SERVICE
135/tcp    open  msrpc
445/tcp    open  microsoft-ds
9080/tcp   open  glrpc

(kali@kali) ~/Desktop
$
```

Figure 7: `nmap -sS 10.0.2.2` — SYN scan showing open ports 135 (msrpc), 445 (microsoft-ds), 9080 (glrpc)

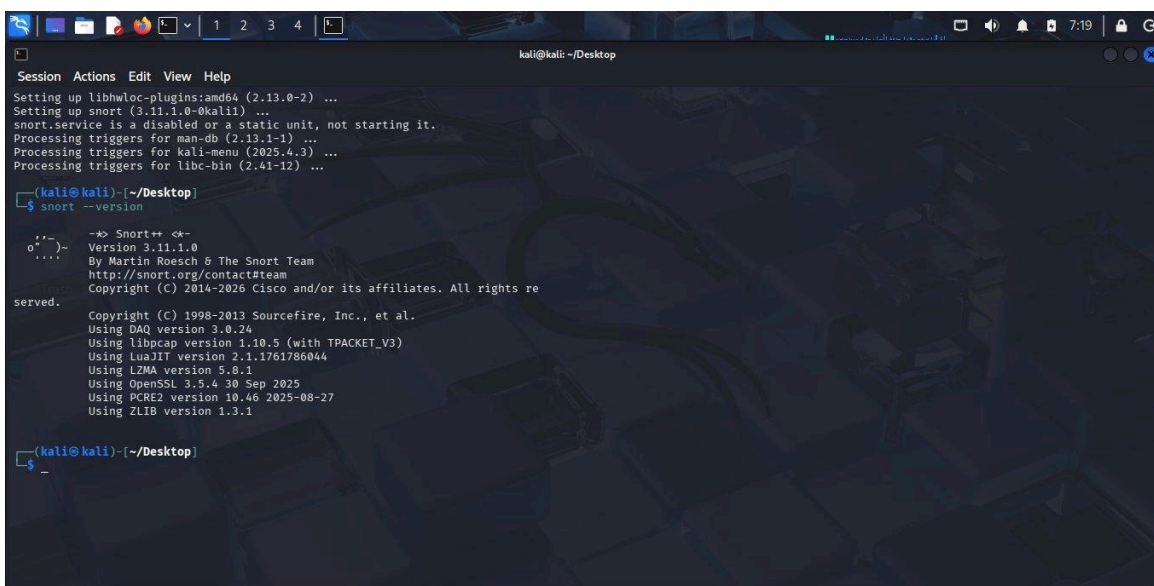
Task 02: Defensive Summary & IDS Implementation

This task involved implementing Snort as an IDS, creating a custom ICMP detection rule, and analyzing attacker footprints from a defensive perspective.

2.1 Snort Installation & Version

```
sudo apt update && sudo apt install snort -y
snort --version
```

Snort 3.11.1.0 was successfully installed on Kali Linux 2025.4.



```
kali@kali: ~/Desktop
Session Actions Edit View Help
Setting up libhwaloc-plugins:amd64 (2.13.0-2) ...
Setting up snort (3.11.1.0-0kali) ...
snort.service is a disabled or a static unit, not starting it.
Processing triggers for man-db (2.13.1-1) ...
Processing triggers for kali-menu (2025.4.3) ...
Processing triggers for libc-bin (2.41-12) ...

(kali@kali)~[~/Desktop]
$ snort --version

--> Snort++ <--
Version 3.11.1.0
By Martin Roesch & The Snort Team
http://snort.org/contact#team
Copyright (C) 2014-2026 Cisco and/or its affiliates. All rights re
served.

Copyright (C) 1998-2013 Sourcefire, Inc., et al.
Using DAQ version 3.0.24
Using libpcap version 1.10.5 (with TPACKET_V3)
Using LuaJIT version 2.1.1761786044
Using LZMA version 5.8.1
Using OpenSSL 3.5.4 30 Sep 2025
Using PCRE2 version 10.46 2025-08-27
Using ZLIB version 1.3.1

(kali@kali)~[~/Desktop]
$
```

Figure 8: `snort --version` — confirming Snort++ Version 3.11.1.0

2.2 ICMP Alert Rule in local.rules

```
sudo nano /etc/snort/rules/local.rules
```

Rule added:

```
alert icmp any any -> any any (msg:"ICMP Packet Detected"; sid:1000001;
rev:1;)
```

Rule breakdown: alert = action | icmp = protocol | any any = source IP/port | -> any any = destination | msg = alert message | sid = unique rule ID | rev = rule revision.

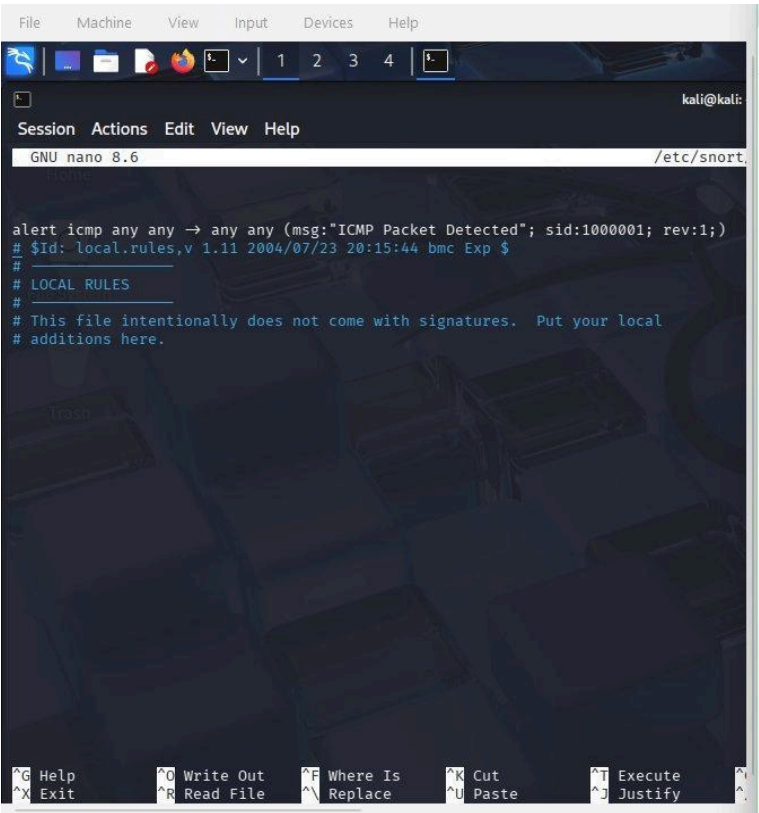


Figure 9: /etc/snort/rules/local.rules — ICMP detection rule

2.3 How is ICMP Different from ARP?

ARP (Address Resolution Protocol) operates at Layer 2 (Data Link Layer) and works only within a local network (LAN). Its purpose is to discover the MAC address associated with a known IP address. ARP cannot cross routers.

ICMP (Internet Control Message Protocol) operates at Layer 3 (Network Layer) and works across different networks including the internet. It is used for testing connectivity (ping), reporting errors, and network diagnostics.

Feature	ICMP	ARP
OSI Layer	Layer 3 (Network)	Layer 2 (Data Link)
Purpose	Connectivity testing & error reporting	Resolves IP to MAC address
Scope	Works across networks (WAN + LAN)	LAN only — cannot cross routers
Example Tool	ping, hping3	arp-scan, netdiscover
IDS Detectable	Yes — caught by Snort rules	Only within LAN segment

2.4 Starting Snort & Sending ICMP Packets

Starting Snort in IDS Mode

```
sudo snort -c /etc/snort/snort.lua -R /etc/snort/rules/local.rules -i eth0
-A alert_fast
```

Sending ICMP Packets from Attacking Machine

```
ping 10.0.2.2
```

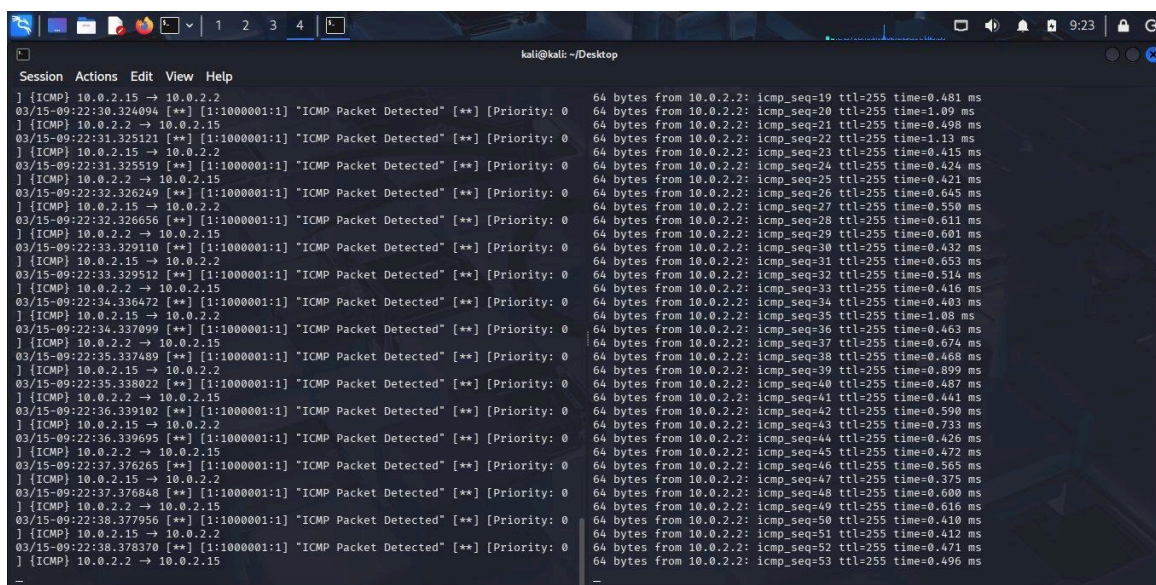


Figure 10: Split screen — LEFT: Snort firing ICMP Packet Detected alerts in real time | RIGHT: ping sending packets

Snort displayed real-time alerts for every ICMP packet detected, logging rule SID 1000001 with source/destination IPs and timestamps.

2.5 Snort Alert Logs

Snort 3 outputs alert logs directly to the console in alert_fast format. Each alert showed:

- **[**] [1:1000001:1] ICMP Packet Detected [**] [Priority: 0]**
- **{ICMP} 10.0.2.15 → 10.0.2.2** (logged with timestamp for each packet)

```

Session Actions Edit View Help
] {ICMP} 10.0.2.15 → 10.0.2.2 64 bytes from 10.0.2.2: icmp_seq=19 ttl=255 time=0.481 ms
03/15-09:22:30.324094 [**] [1:1000001:1] "ICMP Packet Detected" [**] [Priority: 0
] {ICMP} 10.0.2.2 → 10.0.2.15 64 bytes from 10.0.2.2: icmp_seq=20 ttl=255 time=1.09 ms
] {ICMP} 10.0.2.2 → 10.0.2.15 64 bytes from 10.0.2.2: icmp_seq=21 ttl=255 time=0.498 ms
03/15-09:22:31.325121 [**] [1:1000001:1] "ICMP Packet Detected" [**] [Priority: 0
] {ICMP} 10.0.2.15 → 10.0.2.2 64 bytes from 10.0.2.2: icmp_seq=22 ttl=255 time=1.13 ms
03/15-09:22:31.325519 [**] [1:1000001:1] "ICMP Packet Detected" [**] [Priority: 0
] {ICMP} 10.0.2.2 → 10.0.2.15 64 bytes from 10.0.2.2: icmp_seq=23 ttl=255 time=0.415 ms
] {ICMP} 10.0.2.2 → 10.0.2.15 64 bytes from 10.0.2.2: icmp_seq=24 ttl=255 time=0.424 ms
03/15-09:22:32.326249 [**] [1:1000001:1] "ICMP Packet Detected" [**] [Priority: 0
] {ICMP} 10.0.2.15 → 10.0.2.2 64 bytes from 10.0.2.2: icmp_seq=25 ttl=255 time=0.421 ms
] {ICMP} 10.0.2.2 → 10.0.2.15 64 bytes from 10.0.2.2: icmp_seq=26 ttl=255 time=0.645 ms
03/15-09:22:32.326656 [**] [1:1000001:1] "ICMP Packet Detected" [**] [Priority: 0
] {ICMP} 10.0.2.2 → 10.0.2.15 64 bytes from 10.0.2.2: icmp_seq=27 ttl=255 time=0.550 ms
] {ICMP} 10.0.2.15 → 10.0.2.2 64 bytes from 10.0.2.2: icmp_seq=28 ttl=255 time=0.611 ms
03/15-09:22:33.329110 [**] [1:1000001:1] "ICMP Packet Detected" [**] [Priority: 0
] {ICMP} 10.0.2.2 → 10.0.2.15 64 bytes from 10.0.2.2: icmp_seq=29 ttl=255 time=0.601 ms
] {ICMP} 10.0.2.15 → 10.0.2.2 64 bytes from 10.0.2.2: icmp_seq=30 ttl=255 time=0.432 ms
03/15-09:22:33.329512 [**] [1:1000001:1] "ICMP Packet Detected" [**] [Priority: 0
] {ICMP} 10.0.2.2 → 10.0.2.15 64 bytes from 10.0.2.2: icmp_seq=31 ttl=255 time=0.653 ms
] {ICMP} 10.0.2.15 → 10.0.2.2 64 bytes from 10.0.2.2: icmp_seq=32 ttl=255 time=0.514 ms
03/15-09:22:33.336472 [**] [1:1000001:1] "ICMP Packet Detected" [**] [Priority: 0
] {ICMP} 10.0.2.2 → 10.0.2.15 64 bytes from 10.0.2.2: icmp_seq=33 ttl=255 time=0.416 ms
] {ICMP} 10.0.2.15 → 10.0.2.2 64 bytes from 10.0.2.2: icmp_seq=34 ttl=255 time=0.403 ms
03/15-09:22:34.337089 [**] [1:1000001:1] "ICMP Packet Detected" [**] [Priority: 0
] {ICMP} 10.0.2.2 → 10.0.2.15 64 bytes from 10.0.2.2: icmp_seq=35 ttl=255 time=1.08 ms
] {ICMP} 10.0.2.15 → 10.0.2.2 64 bytes from 10.0.2.2: icmp_seq=36 ttl=255 time=0.463 ms
03/15-09:22:34.337489 [**] [1:1000001:1] "ICMP Packet Detected" [**] [Priority: 0
] {ICMP} 10.0.2.2 → 10.0.2.15 64 bytes from 10.0.2.2: icmp_seq=37 ttl=255 time=0.674 ms
] {ICMP} 10.0.2.15 → 10.0.2.2 64 bytes from 10.0.2.2: icmp_seq=38 ttl=255 time=0.468 ms
03/15-09:22:35.338022 [**] [1:1000001:1] "ICMP Packet Detected" [**] [Priority: 0
] {ICMP} 10.0.2.2 → 10.0.2.15 64 bytes from 10.0.2.2: icmp_seq=39 ttl=255 time=0.899 ms
] {ICMP} 10.0.2.15 → 10.0.2.2 64 bytes from 10.0.2.2: icmp_seq=40 ttl=255 time=0.487 ms
03/15-09:22:35.339102 [**] [1:1000001:1] "ICMP Packet Detected" [**] [Priority: 0
] {ICMP} 10.0.2.2 → 10.0.2.15 64 bytes from 10.0.2.2: icmp_seq=41 ttl=255 time=0.441 ms
] {ICMP} 10.0.2.15 → 10.0.2.2 64 bytes from 10.0.2.2: icmp_seq=42 ttl=255 time=0.590 ms
03/15-09:22:36.339695 [**] [1:1000001:1] "ICMP Packet Detected" [**] [Priority: 0
] {ICMP} 10.0.2.2 → 10.0.2.15 64 bytes from 10.0.2.2: icmp_seq=43 ttl=255 time=0.733 ms
] {ICMP} 10.0.2.15 → 10.0.2.2 64 bytes from 10.0.2.2: icmp_seq=44 ttl=255 time=0.426 ms
03/15-09:22:37.376265 [**] [1:1000001:1] "ICMP Packet Detected" [**] [Priority: 0
] {ICMP} 10.0.2.2 → 10.0.2.15 64 bytes from 10.0.2.2: icmp_seq=45 ttl=255 time=0.472 ms
] {ICMP} 10.0.2.15 → 10.0.2.2 64 bytes from 10.0.2.2: icmp_seq=46 ttl=255 time=0.565 ms
03/15-09:22:37.376848 [**] [1:1000001:1] "ICMP Packet Detected" [**] [Priority: 0
] {ICMP} 10.0.2.2 → 10.0.2.15 64 bytes from 10.0.2.2: icmp_seq=47 ttl=255 time=0.375 ms
] {ICMP} 10.0.2.15 → 10.0.2.2 64 bytes from 10.0.2.2: icmp_seq=48 ttl=255 time=0.600 ms
03/15-09:22:37.377056 [**] [1:1000001:1] "ICMP Packet Detected" [**] [Priority: 0
] {ICMP} 10.0.2.2 → 10.0.2.15 64 bytes from 10.0.2.2: icmp_seq=49 ttl=255 time=0.616 ms
] {ICMP} 10.0.2.15 → 10.0.2.2 64 bytes from 10.0.2.2: icmp_seq=50 ttl=255 time=0.410 ms
03/15-09:22:38.378370 [**] [1:1000001:1] "ICMP Packet Detected" [**] [Priority: 0
] {ICMP} 10.0.2.2 → 10.0.2.15 64 bytes from 10.0.2.2: icmp_seq=51 ttl=255 time=0.412 ms
] {ICMP} 10.0.2.15 → 10.0.2.2 64 bytes from 10.0.2.2: icmp_seq=52 ttl=255 time=0.471 ms
] {ICMP} 10.0.2.2 → 10.0.2.15 64 bytes from 10.0.2.2: icmp_seq=53 ttl=255 time=0.496 ms

```

Figure 11: Snort alert log — timestamped ICMP detections confirming rule SID 1000001 firing

2.6 ICMP Flooding Attack & Smurf Attack

ICMP Flooding Attack (DoS)

An ICMP Flood attack is a Denial of Service (DoS) attack where an attacker sends an overwhelming number of ICMP Echo Request packets to a single target. The target's CPU and network bandwidth become exhausted trying to process and respond to each packet, making it unavailable to legitimate users. This is a direct one-to-one attack.

Smurf Attack (DDoS)

A Smurf Attack is a Distributed Denial of Service (DDoS) attack that amplifies the ICMP flood. The attacker spoofs the victim's IP address and sends ICMP Echo Requests to a network's broadcast address. Every device on that network replies to the victim simultaneously — multiplying the traffic by the number of devices. One attacker generates a massive flood using many unwitting devices as amplifiers.

	ICMP Flood (DoS)	Smurf Attack (DDoS)
Attack Type	Direct DoS	Amplified DDoS
Method	Flood target directly with pings	Spoof victim IP, broadcast ping
Amplification	None (1:1)	High (1:many)
Detectability	Easier — single source	Harder — distributed sources
Mitigation	Rate limiting, firewall	Disable broadcast ping, ingress filtering

Conclusion

This Week 02 lab provided hands-on experience with both offensive and defensive network security techniques:

- Passive reconnaissance using whois and DNS enumeration revealed domain infrastructure without touching the target
- Active reconnaissance with Nmap discovered open ports, services, and version information on live hosts
- The TCP SYN Scan demonstrated stealth scanning — leaving fewer traces than full connect scans
- Wireshark packet analysis confirmed the SYN-SYN-ACK-RST pattern characteristic of stealth scanning
- Snort IDS successfully detected ICMP traffic using a custom signature rule, demonstrating signature-based detection

The key learning from this lab: every offensive technique has a corresponding defensive detection method. A SOC analyst monitoring with Snort or Wireshark can identify the same patterns that an attacker uses — making signature-based detection a powerful first line of defense.

— End of Report —